

Nathan Liu

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EDUCATION

University of California, Berkeley

May 2026

Bachelor of Arts in Data Science & Computer Science

GPA: 3.84/4.00

- **Coursework:** Structure and Interpretation of Computer Programs, Data Structures, Machine Structures, Introduction to Data Science, Probability for Data Science, Principles & Techniques of Data Science, Introduction to Full-stack Web Development, Efficient Algorithms and Intractable Problems, Optimization Models in Engineering

PROJECTS

Youtube NLP | *React, Node.js, Firebase, GCP, Gemini API*

Fall 2025

- Developed an AI-powered web app to analyze trending YouTube titles, achieving ~80% accuracy in predicting trending characteristics and ~85% correlation with higher view counts.
- Engineered a serverless backend (Node.js/Express on Firebase Functions) integrating YouTube Data API v3 for data retrieval.
- Leveraged NLP (keyword extraction, frequency analysis) and the Gemini API for advanced title analysis and performance rating.
- Built a responsive React front-end providing users instant, data-driven feedback to optimize video titles.

Personal Portfolio | *HTML/CSS, JavaScript, Node.js, ImageMagick, Wrangler, S3 API, Cloudflare*

Fall 2025

- Built a full-stack personal website with dynamic APIs and image processing using Express.js, Node.js, and ImageMagick.
- Deployed via Cloudflare with GitHub CI/CD and Wrangler for automated builds and deployments.
- Integrated S3 buckets for efficient media storage and retrieval with dynamic uploading to Cloudflare and automatic website update.

SpotifySync | *React.js, Node.js, Express.js, MongoDB, WebSocket*

Summer 2024

- Developed a real-time collaborative playlist builder (Spotify API) featuring a synchronized swipe UI requiring mutual 'likes' for song addition.
- Engineered the full-stack application using WebSockets for live synchronization, React.js (front-end), Node.js/Express.js (back-end), and MongoDB (session data).
- Implemented CI/CD pipelines via GitHub Actions to automate testing and linting, enhancing workflow efficiency.

HappyLemon Inventory Analytics | *MySQL, Python, Pandas, Matplotlib*

Fall 2024

- Designed a centralized SQL database consolidating sales and inventory data for real-time tracking.
- Engineered a data integration pipeline using Python (mysql.connector, Pandas) to aggregate POS system data with stock levels, creating cohesive datasets for visualization with Matplotlib.
- Benefited long-term inventory maximization which created a roughly 15% decrease in spending on inventory and streamlined employee schedules to maximize profit and furthermore minimize spending.

EXPERIENCE

HappyLemon USA

Sept. 2021 – Sept. 2024

Shift Manager

Danville, CA

- Led and scheduled a team of 15+ employees, ensuring operational efficiency and productivity over three years.
- Trained and onboarded 10+ new staff members, optimizing both front and back-of-house workflows.
- Managed daily operations, resolved customer concerns, and fostered a positive team environment through effective communication.

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java, JavaScript, TypeScript, HTML/CSS

Framework & Libraries: React.js, Node.js, Express.js, MongoDB, Pandas, NumPy, Seaborn, WebSockets, PyTorch

Developer Tools: Git, GitHub, Docker, Cloudflare, Firebase, AWS, GCP, Jupyter Notebook, VS Code